



Probability & Statistics Honors

OPTIMA ACADEMY ONLINE · COURSE SYLLABUS · 2026–2027

Full Year · Grades 9–12 · Honors

FACILITATOR

[Facilitator name — click to edit]

FLORIDA COURSE

Florida 1210300 · Full-year honors course · 1 mathematics credit

GRADE LEVEL

Grades 9–12 (High School)

EMAIL & OFFICE

[your email] · 239.673.2801 (OAO Front Office)

OFFICE HOURS

[Days & times — click to edit]

FORMAT

On-Demand (also offered live on the VR campus)

About this syllabus. This is the plan for the course as designed. It is a full-year high-school honors course: two semesters of 17 lesson weeks plus a finals week each, 36 weeks in all. On-Demand scholars pace through the twelve modules across the four quarters of their enrolled year; On-Demand and live-school sections share the same content and expectations, though exact pacing, scheduled dates, and a few activities may vary from section to section and term to term. Your facilitator will confirm the specifics for your section.

COURSE OVERVIEW

Probability & Statistics Honors teaches scholars to reason carefully about chance, evidence, and decisions — and then to use that reasoning on real data. The year is built in two movements. **Semester 1 is probability and decision:** it opens with a game-show puzzle that fooled professors of mathematics, builds the rules of probability from logic upward, confronts the Gambler's Fallacy and the base-rate fallacy, develops random variables and expected value, and closes by asking what probability even *means* — a genuine unresolved argument in the discipline. **Semester 2 is statistics:** honest data displays, descriptive statistics, the normal curve, regression and the discipline of "correlation is not causation," sound sampling and study design, and finally statistical inference — confidence intervals and significance — built by simulation on the probability foundation the first semester laid.

The work is hands-on every week: interactive simulators run right on the lesson pages, every module includes two virtual-reality activities on Optima's VR campus (walking a life-sized probability tree, releasing balls down a room-scale Galton board), every lesson teaches the exact keystrokes for the free Desmos scientific calculator (and, in Semester 2, Excel), and every lesson includes an AI Training task in which scholars put a claim to an AI tool and then *check its answer against their own work*.

WHAT SCHOLARS WILL LEARN TO DO

- Compute and interpret probabilities using sample spaces, the addition and multiplication rules, counting, permutations, and combinations.
- Reason with conditional probability and Bayes' theorem — and diagnose the Gambler's Fallacy and the base-rate fallacy in the wild.
- Model uncertainty with random variables and probability distributions, including the binomial and geometric, and interpret expected value.
- Weigh decisions under risk using expected values, standard deviations, and utility — including why diversification reduces risk.
- Choose, build, and interpret honest data displays — and spot misleading ones in media and advertising.
- Summarize and compare distributions with measures of center, variability, and shape; fit and read the normal model.
- Fit linear and exponential models to bivariate data, interpret correlation, and explain why correlation is not causation.
- Design sound samples, surveys, and randomized experiments; estimate with margins of error and confidence intervals; judge significance by simulation; and critique data-based reports.

THE YEAR AT A GLANCE

Each module runs two to three weeks. A normal week holds three lessons, each paired with one assignment and one five-question quiz. Every module also includes an early VR activity, a culminating VR lab, and short begin/look-back reflections. Week numbers below follow the sample full-year pacing; On-Demand scholars pace to their own enrolled term.

SEMESTER 1 — PROBABILITY & DECISION	SEMESTER 2 — STATISTICS
1. Foundations of Probability & Logic (wks 1–3) 2. Calculating Probabilities & Counting (wks 4–6) 3. Conditional Probability & Bayes (wks 7–9) 4. Random Variables & Distributions (wks 10–12) 5. Decision Theory & Rational Choice (wks 13–15) 6. Foundations of Inference (wks 16–17) Finals Week — review + Semester 1 Final Exam (wk 18)	7. Data & Data Displays (wks 19–21) 8. Descriptive Statistics (wks 22–24) 9. The Normal Distribution (wks 25–26) 10. Bivariate Data & Regression (wks 27–29) 11. Sampling & Study Design (wks 30–32) 12. Inference & Statistical Reasoning (wks 33–35) Finals Week — review + Semester 2 Final Exam (wk 36)

MAJOR MILESTONES

Target dates below follow a sample full-year pacing on the 2026–2027 calendar; On-Demand scholars pace to their own enrolled term and Live sections follow the class schedule, keeping each milestone roughly in order.

MILESTONE	COURSE PLACE	SAMPLE TARGET	WHAT YOU TURN IN
Module VR Labs (12 across the year)	End of every module	every 2–3 weeks	A VR snapshot or short spatial recording plus a brief reflection, submitted in Canvas and scored on a three-part rubric.
Semester 1 Final Exam	Week 18	by Dec 18 (Q2)	A 50-question comprehensive exam. Every question is drawn from the lesson quizzes you have already taken — nothing on it is new.
Excel data labs	Semester 2, periodic	Q3–Q4	Finished Excel workbooks (charts, summary statistics, regression) uploaded to Canvas; every click and formula is taught step by step in the lesson.
Semester 2 Final Exam	Week 36	by May 28 (Q4)	A 50-question comprehensive exam drawn entirely from Semester 2's lesson quizzes.

IMPORTANT DATES — 2026–2027

Aug 12, 2026	First day of school	Sep 7	Holiday (Labor Day)
Oct 9	Quarter 1 ends	Oct 12	Holiday (Columbus Day)
Nov 11	Holiday (Veterans' Day)	Nov 23–27	Thanksgiving break
Dec 18	Quarter 2 ends · fall semester ends	Dec 21–Jan 1	Winter break
Jan 5, 2027	Classes resume	Jan 18	Holiday (MLK Day)
Feb 15	Holiday (Presidents' Day)	Mar 5	Quarter 3 ends
Mar 26–Apr 5	Spring break	Apr 6	Classes resume
May 28, 2027	Quarter 4 ends · last day of school	May 31	Holiday (Memorial Day)

ASSESSMENT & GRADING

50%

Weekly lesson assignments & lesson quizzes

30%

VR labs, VR activities & module reflections

20%

Category weights are confirmed by your facilitator in Canvas and may vary slightly by section. A promise worth knowing now: **every question on each semester final comes from a lesson quiz you have already taken.** Honest weekly work is the exam review.

HOW THIS COURSE IS DELIVERED

This course is offered in two formats that share the same content, assignments, and expectations: an **On-Demand (asynchronous)** section you pace yourself, and a **Live (synchronous)** section that meets on a schedule on Optima's VR campus. The two overviews below explain how each one works. Find the one that matches your enrollment.

ON-DEMAND (ASYNCHRONOUS) LEARNING

In the On-Demand section you move through the course at your own pace within each quarter, guided by due dates that keep you on track. All assignments for a quarter open at its start and close at its end. Because the course is On-Demand, there is no grade penalty for missing an individual due date — but any assignment not completed by the end of the quarter receives a **zero**, and assignments submitted in the final **14 days** of a quarter receive a **10% deduction**, so steady pacing matters.

Attendance is measured by assignment completion. Per school policy, a scholar who completes no assignment for ten consecutive school days, without notifying the school of an absence in advance, may be dropped from the course. Completion means a **sincere attempt** — real effort on real work. Rushed work, wrong or corrupted files, irrelevant or nonsense submissions, AI-generated filler, and word-salad responses do not count.

You are never on your own. A live facilitator grades your work, gives feedback, and hosts weekly office hours, study hall, and VR workshops.

LIVE (SYNCHRONOUS) LEARNING

In the Live section you join scheduled class meetings on Optima's VR campus, where your teacher leads instruction in real time and you learn alongside classmates. The lessons, assignments, quizzes, and exams are the same as the On-Demand course; what differs is the pacing — a Live section moves through the material together on the class schedule, and class time is also used for discussion, worked problems, and guided practice with the course's simulators and VR spaces.

Attendance is measured by presence and participation. You are expected to attend the scheduled live sessions, on time and ready to take part. If you must miss a class, notify the school in advance; your teacher will let you know how to make up the work and any material you missed. As in every Optima course, completing your work means a sincere attempt — real effort on real work — and assignments still open and close with each quarter.

APPROPRIATE USE OF AI

This course does something unusual: it **teaches** you to use AI tools with a statistician's skepticism. Every lesson includes an AI Training task built on one pattern — *prompt, predict, verify*. You craft a question for an AI, work out (or predict) the answer yourself, then check the AI's response against your own reasoning and flag where it is wrong, hand-wavy, or misleading. Probability and statistics are fertile ground for this: AI tools frequently mishandle conditional probability, base rates, and the difference between correlation and causation, and by October you will catch them doing it.

The integrity rule is the same as in every Optima course. You are welcome to use AI tools to **support** your learning — to explain a concept a different way, check your understanding, or generate extra practice — unless a particular assignment says otherwise. What AI may never do is the thinking or the work an assignment asks of you: your tree diagrams, computations, verdicts, reflections, and Excel workbooks must be genuinely your own. Submitting AI-generated work as your own is dishonest and counts as an insincere attempt. When you do use an AI tool on an assignment, say so briefly. The goal of the whole year is to be the person who can judge whether an AI's statistical claim is any good — which you can only do if you can do the work without it.

LEARNING IN VIRTUAL REALITY

Optima is a virtual-reality school, and in this course VR is where probability becomes physical. Every module includes two solo VR assignments on Optima's VR campus: an early hands-on activity and a culminating lab — you will arrange all 36 outcomes of a two-dice roll on a floor grid, walk a life-sized probability tree, stand in the "tested positive" corridor of a thousand-person screening ward, and watch a room-scale Galton board pour randomness into a bell curve. Each session runs about 20–30 minutes and ends with a snapshot or short recording submitted in Canvas.

Keep your headset charged and ready, and treat the VR campus with the same courtesy and seriousness you would a physical one. If you ever cannot get into VR for a session, tell your facilitator — there is an alternative way to take part so you do not fall behind.

BEING AN OPTIMA SCHOLAR

Optima's mission is an excellent, virtue-centered education, and virtue is at the center of this course as much as any other. Every scholar is expected to practice the seven virtues that shape life at Optima — **courage, courtesy, honesty, perseverance, self-government, service, and responsibility**. This course gives each module its own virtue thread, and the mathematics earns it: honesty is what Bayes' theorem asks of you when the evidence says your belief was wrong; responsibility is what "correlation is not causation" demands before you make a claim; courage is what deciding under genuine risk requires.

In practice: do your own honest work; treat everyone you meet in, through, or because of this course with respect — in writing, in discussion, and in VR; engage actively rather than passively; come to office hours to get help, meet other scholars, and get to know your facilitator; and take responsibility for your own pacing and progress. You belong to a house and a community of scholars, not only to a course.

WHAT YOU NEED & STAYING CONNECTED

- ✓ A computer with reliable internet and access to **Canvas** — every lesson, simulator, assignment, and quiz lives there.
- ✓ A **VR headset**, charged and ready for each module's two immersive activities on Optima's VR campus.
- ✓ The free **Desmos Scientific Calculator** at desmos.com/scientific — nothing to buy or install; every lesson teaches the exact buttons.
- ✓ **Microsoft 365 (Excel)** with OneDrive, for the Semester 2 data labs and workbook submissions.
- ✓ A notebook or graph-ruled binder and pencils — tree diagrams, tables, and worked problems are drawn by hand all year.

Staying connected. Updates about this course are posted on the course's Canvas page; school-wide updates are posted in your Homeroom Canvas course. Check both regularly. Your facilitator holds office hours and is glad to help with the material, to meet you, or to talk through your plans — come by, or reach out any time by email.

Probability & Statistics Honors · Florida course 1210300 · Full-year high-school honors course (1 mathematics credit) · Optima Academy Online, 2026–2027. Built from the course content (12 modules / 34 lesson weeks + 2 finals weeks) and the board-approved 2026–2027 academic calendar. Sample target dates reflect full-year pacing; On-Demand scholars pace to their own enrolled term. School calendar dates may be adjusted by the school.